Bayesian Method

Xiao-Hua Zhou

Peking University

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Frequentist versus Bayesian Inference

- Frequentist (or classical) methods, sometimes called procedural inference, interpreted probabilities as long run frequencies.
- The goal of frequentist inference is to create procedures with long run guarantees; here the guarantees should be uniform over θ if possible.
- For example, a confidence interval covers the true value of θ with probability $1 - \alpha$, no matter what the true value of θ is.
- In frequentist inference, procedures are random while parameters are fixed, unknown quantities.

Frequentist versus Bayesian Inference

- The Bayesian approach regarded probability as a measure of subjective degree of belief.
- In this framework, everything, including parameters, is regarded as random, and there are no long run frequency guarantees.

Frequentist versus Bayesian Inference

	Bayesian.	Frequentist
Probability	subjective degree of belief	limiting frequency
Goal	analyze beliefs	create procedures with frequency guarantees
θ	random variable	fixed
X	random variable	random variable
Use Bayes' theorem?	Yes. To update beliefs.	Yes, if it leads to procedure with good frequentist behavior. Otherwise no.

Bayesian Inference

- Unlike the likelihood method, Bayesian analysis treats θ as a random variable.
- Let $X_1, \ldots, X_n \sim p(x \mid \theta)$.
- Let $p(\theta)$ be a prior distribution on θ.

Bayes Estimator

- Note that the joint distribution of $X_1, \ldots, X_n, \theta$ can be written as follows:

$$p(x_1, \ldots, x_n, \theta) = p(x_1, \ldots, x_n \mid \theta) p(\theta).$$

Bayes Estimator

- Using Bayes' theorem, we obtain the following posterior distribution on θ:

$$p(\theta \mid x_1, \ldots, x_n) = \frac{p(x_1, \ldots, x_n \mid \theta) p(\theta)}{p(x_1, \ldots, x_n)},$$

where

$$p(x_1, \ldots, x_n) \int p(x_1, \ldots, x_n \mid \theta) p(\theta) d\theta.$$

- This can be written as

$$p(\theta \mid x_1, \ldots, x_n) \approx L(\theta) p(\theta) = \text{ Likelihood } \times \text{ prior.}$$

Bayes Estimator

- A Bayes estimator is based on the posterior distribution.
- One particular Bayes estimator is the mean of the posterior distribution:

$$\widehat{\theta} = E(\theta \mid x_1, \ldots, x_n) = \int \theta p(\theta \mid x_1, \ldots, x_n) d\theta = \frac{\int \theta p(x_1, \ldots, x_n \mid \theta) p(\theta) d\theta}{\int p(x_1, \ldots, x_n \mid \theta) p(\theta) d\theta}.$$

Example 7

- Let $X_1, \ldots, X_n \sim \textit{Bernoulli}(\theta)$.
- Let the prior on θ be $\theta \sim \textit{Beta}(\alpha, \beta)$. That is,

$$p(\theta) = \theta^{\alpha-1}(1-\theta)^{\beta-1}/(\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}),$$

  where $\Gamma(\alpha) = \int_0^\infty t^{\alpha-1}e^{-t}dt$.

- Let $Y = \sum_i X_i$. Then,

$$\underbrace{\theta^{Y}(1-\theta)^{n-Y}}_{\text{Likelihood}} \times \underbrace{\theta^{\alpha-1}(1-\theta)^{\beta-1}}_{\text{Prior}} \approx \theta^{Y+\alpha-1}(1-\theta)^{n-Y+\beta-1}$$

How Does Prior Come from?

Two schools of thought on the prior:

Subjective Bayesian:

- The prior is a summary of our subjective beliefs about the data. E.g., in the coin flipping example: the prior for θ should be strongly peaked around $1/2$.

Objective Bayesian

- The prior should be chosen in a way that is "uninformed".
- E.g., in the coin flipping example: the prior should be uniform on $[0, 1]$.

How Does Prior Come from?

- Objective Bayesian inference was a response to the basic criticism that subjectivity should not enter into scientific conclusions.
- A flat prior is a uniform prior on the parameter: $f(\theta)$ is constant.
- The flat prior may not be improper: it is not a probability distribution on its own, but yields a well-defined posterior.
- Flat priors can held up as evidence of why Bayesian estimates are at least as informative as frequentist estimates, since at worst by using a flat prior we can recover maximum likelihood estimation.

Here Does Prior Come from?

- The most conservative thing to do is to assume you have no knowledge, except what is in the data; this is what the flat prior is meant to be
- But flat priors can also lead to some unusual behavior. For example, suppose we place a flat prior on μ ($N(\mu, \sigma^2)$). What is our prior on, μ^3:

$$f(\mu^3) = f(\mu)/(3\mu^2) = 1/(3\mu^2),$$

  which is not flat
- The flat priors are not invariant to transformations of the parameter.

Jeffreys' priors

- Jeffreys showed that a reasonble uninformative prior that is also transformation invariant is obtained by setting $f(\theta)$ to the inverse of the Fisher information at θ:

$$\pi_J(\theta) \approx \mid det(I(\theta)) \mid^{1/2},$$

where

$$I(\theta) = -E_\theta(\frac{\partial^2 logp(X \mid \theta)}{\partial \theta^2}).$$

Here $\det(I(\theta))$ is the determinant of $I(\theta)$.

Example on Jeffreys' priors

- $X \sim Binomial(n, \theta)$, where $0 \leq \theta \leq 1$.
- the density function,

$$p(x \mid \theta) = \binom{n}{x} \theta^x (1-\theta)^{n-x},$$

$$logp(x \mid \theta) = xlog(\theta) + (n-x)log(1-\theta).$$

- $$\frac{\partial^2 p(x \mid \theta)}{\partial \theta^2} = -\frac{x}{\theta^2} - \frac{n-x}{(1-\theta)^2}$$

$$I(\theta) = \frac{n}{\theta(1-\theta)}.$$

Hence $\pi_J(\theta) \approx \theta^{-1/2}(1-\theta)^{-1/2}$, is the form of a $Beta(\frac{1}{2}, \frac{1}{2})$ density.

Limitations on Jeffreys' priors and Reference Priors

(Robert, C., The Bayesian Choice (2nd Edition), Springer, 2001, Michael Jordan)

- For one-dimensional parameter, Jeffreys priors work well.
- For vector of multiple parameters, Jeffreys priors may not work well.
- Example: Let $X \sim N(\mu, \sigma^2)$; here $\theta = (\mu, \sigma^2)$. The Fisher information matrix is:

$$I(\theta) = \begin{pmatrix} \frac{1}{\sigma^2} & 0 \\ 0 & \frac{1}{\sigma^2} \end{pmatrix}$$

Hence, the Jeffrey' prior is: $\pi_J(\theta) \approx \mid det(I(\theta)) \mid^{1/2} = \frac{1}{\sigma^2}$.

Limitations on Jeffreys' priors and Reference Priors

- Reference Priors: maximizing some measure of distance or divergence between the posterior and prior, as data observations are made.
- Some divergence measures: Kullback-Leibler divergence or the Hellinger distance.
- By maximizing the divergence, the prior allows the data to have the maximum effect on the posterior estimates.

Reference Priors

- Let $X_1, \ldots, X_n \sim p(. \mid \theta)$, the sample. Let T be the sufficient statistics, and $p(t \mid \theta)$ be the density function of T.
- Reference prior is the uninformative prior that maximizes the divergence between prior and posterior, given T:

$$p(\theta) = argmax_{\pi(\theta)} I(\theta, T),$$

where $I(\theta, T)$ is the expected KL-divergence between the posterior and the prior:

$I(\theta, T) = \int \pi(\theta) \int p(\theta \mid t) \log \frac{p(\theta|t)}{p(\theta)} d\theta dt.$

Note that the K-L divergence is $\int p(\theta \mid t) \log \frac{p(\theta|t)}{p(\theta)} d\theta$

Reference Priors

- When θ is one-dimensional, the reference prior is identical to the Jeffrey' s prior (Bonus problem).

Conjugate Prior

- Let $\mathcal{F}$ denote the class of pdfs or pmfs.
- A class of $\mathcal{T}$ of prior distributions conjugate family for $\mathcal{F}$ if the posterior distribution is in the class $\mathcal{T}$

Example 8

- Let $X_1, \ldots, X_n \sim N(\mu, \sigma^2)$, where σ^2 is known.
- Let $\mu \sim N(m, \tau^2)$. Then
- Therefore, we can show that the posterior distribution of θ is also normal, with mean and variance below:

$$E(\mu \mid X) = \frac{\tau^2}{\tau^2 + \sigma^2/n}\bar{X} + \frac{\sigma^2/n}{\tau^2 + \sigma^2/n}m,$$

$$Var(\mu \mid X) = \frac{\sigma^2\tau^2/n}{\tau^2 + \sigma^2/n}$$

- The normal family is its own conjugate family.

Example 8, continued

- The Bayesian is a linear combination of the prior and sample means.
- If the prior variance τ^2 tends to infinity, the Bayes estimator tends to the sample mean. We can interpret this as saying that, as the prior information becomes more vague, the Bayesian estimator tends to give more weight to the sample information.
- If the prior information is good (that is, $\sigma^2 > \tau^2$), more weight is given to the prior mean.

Example 9

- Suppose we have n_1 control patients and n_2 treatment patients and that X_1 control patients survive while X_2 treatment patient survive.
- We want to estimate $\tau = p_1 - p_2$.
- $X_1 \sim \mathit{Binomial}(n_1, p_1)$ and $X_2 \sim \mathit{Binomial}(n_2, p_2)$.
- Suppose we take the prior as $f(p_1, p_2) = 1$. Then, the posterior is

$$f(\theta_1, \theta_2 \mid x_1, x_2) \approx p_1^{x_1}(1-p_1)^{n_1-x_1} p_2^{x_2}(1-p_2)^{n_2-x_2}.$$

Example 9

- Note that (p_1, p_2) live on a rectangle (a square, actually) and $f(p_1, p_2 \mid x_1, x_2) = f(p_1 \mid x_1) f(p_2 \mid x_2)$.
- where

  $f(p_1 \mid x_1) = p_1^{x_1}(1-p_1)^{n_1-x_1}$, $f(p_2 \mid x_1) = p_2^{x_2}(1-p_2)^{n_2-x_2}$
- This means that p_1 and p_2 are independent under the posterior distribution.
- If we simulate $P_{1,1}, \ldots, P_{1,B} \sim Beta(x_1 + 1, n_1 - x_1 + 1)$ and $P_{2,1}, \ldots, P_{2,B} \sim Beta(x_2 + 1, n_2 - x_2 + 1)$, $\tau_b = P_{2,b} - P_{1,b}$, $b = 1, \ldots, B$, is a sample from $f(\tau \mid x_1, x_2)$.

Frequentist Versus Bayesian Method

- In general, Bayesian and frequentist inferences can be quite different.
- If C is a $1-\alpha$ Bayesian interval then

$$P(\theta \in C \mid X_1, \ldots, X_n) = 1 - \alpha$$

- This does not imply that

$$\text{Frequentist cover } = inf_\theta P_\theta(\theta \in C) = 1 - \alpha$$

- Typically, a -1α Bayesian interval has coverage lower than $1-\alpha$. interval then

Frequentist

- In the frequentist approach, probability is interpreted as long run frequencies.
- The goal of frequentist inference is to create procedures with long run guarantees. Indeed, a better name for frequentist inference might be procedural inference. Moreover, the guarantees should be uniform over θ.
- if possible. For example, a confidence interval traps the true value of θ with probability $1 - \alpha$, no matter what the true value of θ is.

Strength and Weakness of Bayesian Inference

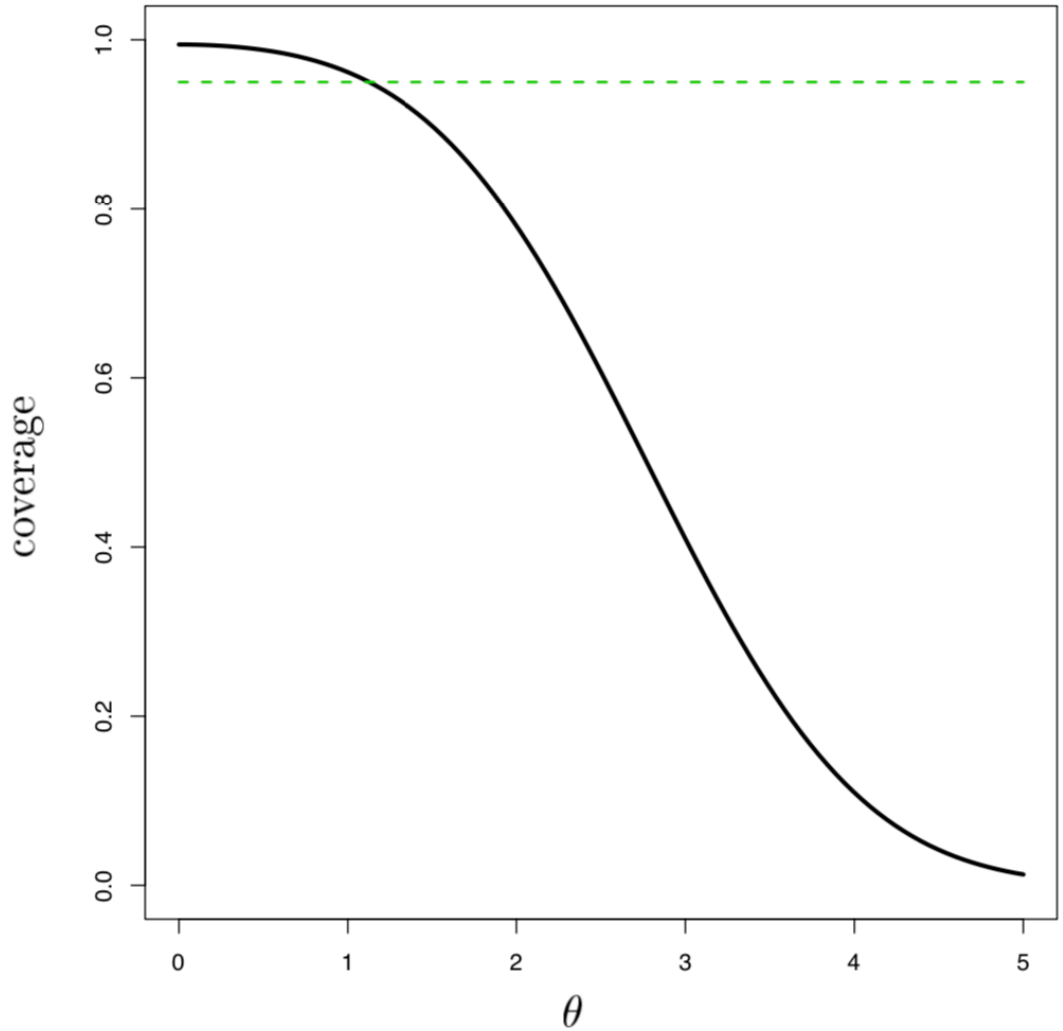
- Bayesian inference is appealing when prior information is available since Bayes" theorem is a natural way to combine prior information with data.
- Bayesian inference allows us to make probability statement about parameters, while frequentist inference provides conference sets C_n which trap the parameter 95 percent of the time.
- In parameter models, with large samples, Bayesian and frequentist methods give approximately the same inference. In general, they do not agree.

Strength and Weakness of Bayesian Inference

- Let $X \sim N(\theta, 1)$ and suppose that we use the prior $\theta \sim N(0, \tau^2)$. We can easily show that the posterior is $\theta \mid x \sim N(cx, c)$, where $c = \frac{1}{1+1/\tau^2}$
- A $1-\alpha$ percent posterior interval is $C = (a, b)$, where $a = cx - \sqrt{c}z_{1-\alpha/2}$ and $b = cx + \sqrt{c}z_{1-\alpha/2}$. $P(\theta \in C \mid X) = 1 - \alpha$.
- We can now ask, from a frequentist perspective, what is the coverage of C, that is, how often will this interval contain the true value?
- The answer is: $P_\theta(a < \theta < b) = P(\theta > a) - P(\theta > b) = \Phi(\frac{\theta(1-c)+z_{1-\alpha/2}\sqrt{c}}{c}) - \Phi(\frac{\theta(1-c)-z_{1-\alpha/2}\sqrt{c}}{c})$.

Frequentist Coverage of 95% Bayesian Interval

$(\tau = 1, \alpha = 0.05)$



- This result shows that unless the true value of θ is close to 0, the coverage can be very small.
- Thus, upon repeated use, the Bayesian 95% confidence interval might contain the true value with frequency near 0.
- In contrast, a frequentist conference interval has coverage 95% coverage no matter what the true value of θ

- This example illustrates the Frequentist and Bayesian methods can be different, but they are answering the different questions.
- Combine prior beliefs with data in a principled way, use bayesian inference.
- To construct procedures with guaranteed long run performance, such as confidence intervals, use frequentist methods.

Bayesian Tests

- The Bayesian paradigm combines the sample information with the prior information, $\pi(\theta)$ using Bayes' Theorem to obtain the posterior distribution $\pi(\theta \mid x)$. All inference about θ should be based on the posterior distribution.
- In a hypothesis testing problem, the posterior distribution may be used to calculate the probabilities that H_0 and H_1 are true. That is, we compute the posterior probabilities $P(\theta \in \Theta_0 \mid x) = P(H_0 \text{ is true } \mid x)$ and $P(\theta \in \Theta_1 \mid x) = P(H_1 \text{ is true } \mid x)$.
- A Bayesian hypothesis test: Accept H_0 if $P(\theta \in \Theta_0 \mid x) \geq P(\theta \in \Theta_1 \mid x)$ and to reject H_0 otherwise.

Example

- Let $X_1, \ldots, X_n \sim N(\theta, \sigma^2)$. Let the prior distribution for θ is $N(\mu, \tau^2)$, where σ^2, μ, and τ^2 are known.
- We are interested in testing $H_0 : \theta \leq \theta_0$ versus $H_1 : \theta > \theta_0$.
- It is easy to show that the posterior $\pi(\theta \mid \bar{x})$ is normal with mean $(n\tau^2\bar{x} + \sigma^2\mu)/(n\tau^2 + \sigma^2)$ and variance $\sigma^2\tau^2/(n\tau^2 + \sigma^2)$.

Example

- $$P(\theta \in \Theta_0 \mid X) = P(\theta \leq \theta_0 \mid x) = \Phi\left(\frac{\theta_0 - \frac{n\tau^2\bar{X}+\sigma^2\mu}{n\tau^2+\sigma^2}}{\frac{\sigma\tau}{\sqrt{n\tau^2+\sigma^2}}}\right)$$

$$P(\theta \in \Theta_1 \mid X) = P(\theta > \theta_0 \mid x) = 1 - \Phi\left(\frac{\theta_0 - \frac{n\tau^2\bar{X}+\sigma^2\mu}{n\tau^2+\sigma^2}}{\frac{\sigma\tau}{\sqrt{n\tau^2+\sigma^2}}}\right)$$

$$= \Phi\left(-\frac{\theta_0 - \frac{n\tau^2\bar{X}+\sigma^2\mu}{n\tau^2+\sigma^2}}{\frac{\sigma\tau}{\sqrt{n\tau^2+\sigma^2}}}\right)$$

- We accept H_0 if

$$\bar{X} \leq \theta_0 + \frac{\sigma^2(\theta_0 - \mu)}{n\tau^2}$$

Bayesian Intervals

- To keep the distinction between Bayesian and classical sets clear, we refer the Bayesian set estimates as credible sets rather than confidence sets.
- Let $\pi(\theta \mid x)$ be the posterior distribution of θ given $X = x$. Then, for any set $A \subset \Theta$, the credible probability of A is

$$P(\theta \in A \mid x) = \int_A \pi(\theta \mid x) d\theta,$$

and A is a credible set for θ.

Example

- Let $X_1, \ldots, X_n$ be iid $N(\theta, \sigma^2)$ with σ known. Let θ have the prior pdf $N(\mu, \tau^2)$, where μ and τ are known.
- Then, we can show the following posterior distribution result:

$$\pi(\theta \mid \bar{x}) \sim N(\delta^B(\bar{x}), Var(\theta \mid \bar{x})),$$

where

$$\delta^B(\bar{x}) = \frac{\sigma^2}{\sigma^2 + n\tau^2} + \frac{n\tau^2}{\sigma^2 + n\tau^2}\bar{x}$$

and

$$Var(\theta \mid \bar{x}) = \frac{\sigma^2\tau^2}{\sigma^2\tau^2}\sigma^2 + n\tau^2.$$

Example

- Therefore,

$$\frac{\theta - \delta^B(\bar{x})}{\sqrt{Var(\theta \mid \bar{x})}} \sim N(0,1).$$

- Then,

$$P(\delta^B(\bar{x}) - z_{1-\alpha/2}\sqrt{Var(\theta \mid \bar{x})} \leq \theta \leq \delta^B(\bar{x}) + z_{1-\alpha/2}\sqrt{Var(\theta \mid \bar{x})} \mid \bar{x}) = 1 - \alpha$$

Bayesian Computing

- If $\theta = (\theta_1, \ldots, \theta_p)$ is a vector, then the posterior $f(\theta \mid X_1, \ldots, X_n)$ is a multivariate distribution.
- If you are interested in one parameter, θ_2, then you compute the marginal posterior:

$$\pi(\theta_2 \mid X_1, \ldots, X_n) = \int \pi(\theta_1, \ldots, \theta_p \mid X_1, \ldots, X_n) d\theta_1 d\theta_3 \ldots d\theta_p$$

- Usually, this integral is intractable. In practice, we can use Monte Carlo methods. In general,

$$\int f(z)p(z)dz \approx \frac{1}{N}\sum_{i=1}^{N} f(z_i),$$

where z_i's are independently drawn from $p(.)$. Here $p(.)$ is usually the density function.

Bayesian Computing

- Problem: It is hard to draw exact samples from $p(.)$.
- Some sampling methods: Rejection Sampling, Importance Sampling, Markov Chains, Metropolis-Hasting Algorithm, Gibbs Sampler

Rejection sampling

We like to draw a single random draw from a density $p(\theta \mid y)$.

Assume that we have a positive function $g(\theta)$:

- Draw θ at random from the probability density proportional to $g(\theta)$.
- With probability $\frac{p(\theta|y)}{Mg(\theta)}$, accept θ as a draw from $p(.)$. If the draw θ is rejected, return the previous step. Here, M is a constant, satisfying $\frac{p(\theta|y)}{g(\theta)} \leq M$ for all θ.
- An accepted θ has the correct distribution, $p(\theta \mid y)$.

Importance sampling

Suppose that we are interested in $E(h(\theta) \mid y)$, but we cannot generate random draws of θ from $p(\theta \mid y)$.

- We can draw S values, $\theta^1, \ldots, \theta^S$ from the probability density proportional to $g(\theta)$.
- Let $q(\theta \mid y)$ be proportional to $g(\theta \mid y)$. Since $E(h(\theta \mid y)) = \int h(\theta) p(\theta \mid y) d\theta = \frac{\int [h(\theta) q(\theta|y)/g(\theta)] g(\theta) d\theta}{\int [q(\theta|y)/g(\theta)] g(\theta) d\theta}$,E(h($\theta \mid y$)) can be estimated by $\frac{\frac{1}{S}\sum_{s=1}^{S} h(\theta^s) w(\theta^s)}{\frac{1}{S}\sum_{s=1}^{S} w(\theta^s)}$, where $w(\theta^s) = \frac{q(\theta^s|y)}{g(\theta^s)}$.

Markov Chain Monte Carlo (MCMC)

- MCMC: a powerful statistical method for sampling from complex probability distributions by simulating a Markov chain whose equilibrium distribution is the desired distribution, generating samples that approximate the target distribution.

Gibbs

Gibbs sampler: wanting to draw from the posterior distribution of a vector $\theta = (\theta_1, \ldots, \theta_d)$.

- At each iteration t, an ordering of the d subvectors of θ is chosen and each θ_j^t is sampled from the conditional distribution given all the other components of θ: $p(\theta_j \mid \theta_{-j}^{t-1}, y)$, where $\theta_{-j}^{t-1} = (\theta_1^t, \ldots, \theta_{j-1}^t, \theta_{j+1}^{t-1}, \ldots, \theta_d^{t-1})$.

Markov Chain Monte Carlo (MCMC): Metropolis

Draw a starting point θ^0 from a starting distribution $p_0(\theta)$.

For $t = 1, \ldots$:

- (a) Sample a proposal θ^* from a jumping distribution at t, $J_t(\theta^* \mid \theta^{t-1})$. For the Metropolis algorithm, the jumping distribution must be symmetric, satisfying the condition $J_t(\theta_b \mid \theta_a) = J_t(\theta_b \mid \theta_a)$ for all θ_a,θ_b, and t.
- (b) Compute $r = p(\theta^* \mid y)/p(\theta^{t-1} \mid y)$
- (c) Set $\theta^t = \theta^*$ with probability $min(r, 1)$, and $\theta^t = \theta^{t-1}$, otherwise.

Markov Chain Monte Carlo (MCMC): Metropolis

- Given the current value θ^{t-1}, the transition distribution $T_t(\theta^t \mid \theta^{t-1})$ of the Markov chain is a mixture of a point mass at $\theta^t = \theta^{t-1}$, and a weighted version of the jumping distribution, $J_t(\theta^t \mid \theta^{t-1})$, that adjusts for the acceptance rate.
- The above algorithm requires the ability to calculate the ratio r for all (θ, θ^*), draw θ from the jumping distribution $J_t(\theta^* \mid \theta)$ for all θ and t.
- The Metropolis-Hastings algorithm generalizes the above Metropolis algorithm by not requiring the sumping rules J_t to be symmetric.